Module 03 – Production Modeling

EDA

Row Labels	Average of capacity	Average of production_cost	Average of demand
1	496	48	624
2	404	48	396
3	545	56	110
4	453	46	509
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Grand Total	**475**	**50**	**410**

This chart shows the average demand, average capacity and the cost per quarter.

Year	Average of capacity	Average of production_cost	Average of demand
2000	496	$ 52	465
2001	469	$ 51	411
2002	479	$ 47	388
2003	496	$ 45	439
2004	496	$ 51	495
2005	442	$ 49	421
2006	478	$ 44	545
2007	489	$ 50	455
2008	492	$ 47	351
2009	460	$ 48	397
2010	501	$ 51	445
2011	465	$ 49	404
2012	477	$ 48	410
2013	485	$ 47	417
2014	510	$ 49	444
2015	450	$ 51	405
2016	454	$ 44	398
2017	447	$ 55	379
2018	487	$ 54	269
2019	486	$ 51	307
2020	488	$ 51	367
2021	431	$ 55	379
2022	439	$ 49	492
2023	471	$ 50	352
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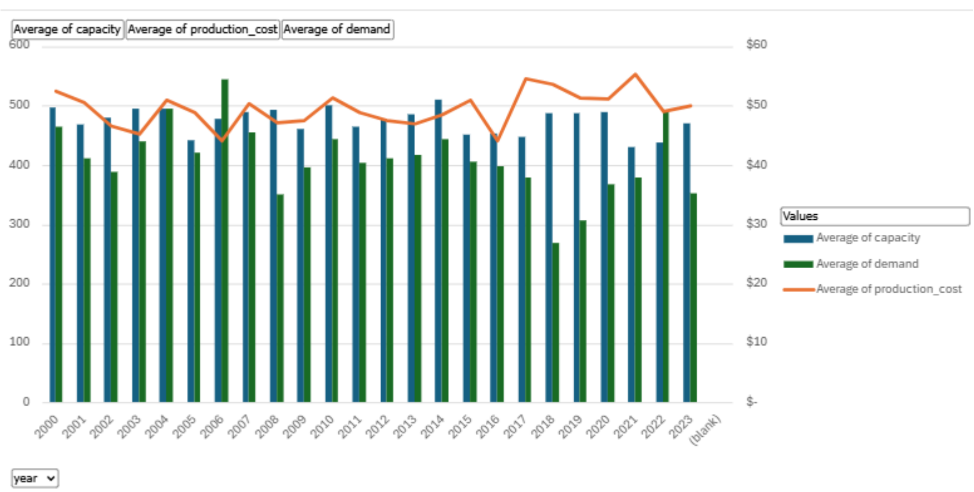
Grand Total	**475**	**$ 50**	**410**

This chart shows the average demand, average capacity and the cost per year.



Yearly chart of the metrics over time

Model Formulation
MIN: 48P1+ 48P2 + 56P3 + 46P4 + 81(B1+B2) /2+ 81(B2+B3)/2+94(B3+B4)/2+ 77(B4+B5)/2

Subject to:
496<=P1<=484 production level month 1
404<=P2<=404 production level month 2
545<= P3 <=545 production level month 3
453<=P4<=57 production level month 4

Where:
B2 = B1+P1-624
B3 = B2 +P2-396
B4 = B3+P3 – 110
B5 = B4+P4 - 509

Model Optimized for Cost Reduction

	Month	1	2	3	4
Beginning Inventory		200	60	68	503
Units Produced		484	404	545	57
Units Demanded		624	396	110	509
Ending Inventory		60	68	503	51
Maximum Production		496	404	545	453
Minimum Inventory		60	40	11	51
Average Inventory		130	64	285.5	277
Unit Production Cost		$ 48	$ 48	$ 56	$ 46
Unit Carrying Cost		$ 81	$ 81	$ 94	$ 77
Monthly Production Cost		$ 23,232	$ 19,392	$ 30,520	$ 2,622
Monthly Carrying Cost		$ 10,483	$ 5,161	$ 26,860	$ 21,407
Total Cost	$ 139,677				

My production and inventory model for the upcoming four months is designed to meet customer demand while minimizing total operational costs. By strategically

adjusting production levels and carefully managing inventory, we aim to keep total costs at $139,677 over the planning horizon.

Key Insights

- **The model ensures production closely aligns with monthly demand while respecting production capacity limits.**
- **Production remains within monthly capacity limits, with peak utilization in Month 3 (545 units produced out of a 545-unit capacity).**
- **Ending inventories are maintained above minimum levels to avoid stockouts, ensuring smooth operations and customer satisfaction.**

Model with Stipulation:

With no production capacity constraint, no units were produced in any month resulting in a continuous declining inventory which resulted in stockouts. Removing carrying costs has changed the cost structure to only reflect lost sales or potential shortages since the carrying cost previously penalized excess inventory. The total cost deficit represents a penalty for negative inventory.

	Month	1	2	3	4
Beginning Inventory		200	-424	-820	-930
Units Produced		0	0	0	0
Units Demanded		624	396	110	509
Ending Inventory		-424	-820	-930	-1439
Maximum Production		496	404	545	453
Minimum Inventory		60	40	11	51
Average Inventory		-112	-622	-875	-1184.5
Unit Production Cost		$ 48	$ 48	$ 56	$ 46
Unit Carrying Cost		$ 81	$ 81	$ 94	$ 77
Monthly Production Cost		$ -	$ -	$ -	$ -
Monthly Carrying Cost		$ (9,032)	$ (50,158)	$ (82,320)	$ (91,538)
Total Cost	$ (233,048)				